HIGH-PROBABILITY MATHEMATICS MOCK TEST

Set 1 • Boards Dominator Series

Time Limit: 2 Hours | **Max Marks:** 50 | **Format:** High-Repeat Templates

SECTION A: Most Repeated MCQs (1 Mark Each)

Q1. Order & Degree: Find the sum of the order and the degree of the differential equation: $\frac{d}{dx}\left[\left(\frac{dy}{dx}\right)^3\right] = 0$ or $\left[1 + \left(\frac{dy}{dx}\right)^2\right]^3 = \frac{d^2y}{dx^2}$.

Q2. Matrix Property: If A is a square matrix of order 3 and |A| = 5, find the value of $|\mathrm{adj}\ A|$.

Q3. Inverse Trigonometry: Find the principal value of $\cot^{-1}(-1/\sqrt{3})$ or $\tan^{-1}(-\sqrt{3})$.

Q4. Vector Properties: If |a + b| = |a - b|, then what is the relationship between vectors a and b?

SECTION B: The "Sure-Shot" Long Answers (5 Marks Each)

Q5. Matrices: System of Linear Equations

Option A: Solve using matrix method: $x - 2y = 10$; $2x - y - z = 8$; $-2y + z = 7$.

Option B: Given A = [[1, 2, -3],[2, 3, 2],[3, -3, -4]] and B = [[-6, 17, 13],[14, 5, -8],[-15, 9, -1]]. Find AB. Hence solve: $x + 2y - 3z = -4$; $2x + 3y + 2z = 2$; $3x - 3y - 4z = 11$.

Q6. 3D Geometry: Shortest Distance

Find the shortest distance between the lines:

$r = (i + 2j - 4k) + \lambda(2i + 3j + 6k)$

$r = (3i + 3j - 5k) + \mu(4i + 6j + 12k)$

Q7. Integrals: Area Under the Curve

Using integration, find the area of the region bounded by the ellipse $x^2/16 + y^2/4 = 1$ between the lines $x = -2$ and $x = 2$.

Q8. Linear Programming (LPP)

Solve graphically: Maximize $Z = 60x + 40y$ Subject to: $x + 2y \leq 12$, $2x + y \leq 12$, $4x + 5y \geq 20$; $x,y \geq 0$.

SECTION C: High-Frequency Short Answers (3 Marks Each)

Q9. Differential Equations (Linear Form): Find the particular solution of $dy/dx - 2xy = 3x^2 e^{(x^2)}$ given $y(0) = 5$.

Q10. Integrals (Properties): Evaluate: $\int_0^{\alpha/2} [1 / (1 + e^{(\sin x)})]\, dx$ OR $\int_0^{\alpha/2} [\sin x / (\sin x + \cos x)]\, dx$.

SECTION D: Case Studies (4 Marks Each)

Q11. Probability Case Study: Bayes' Theorem

Airplane observes severe, moderate, light turbulence (equal probability 1/3). Prob. of late arrival: Severe(55%), Moderate(37%), Light(17%).

1. Find Total Probability of reaching late.
2. If reached late, find prob. it was Moderate turbulence.

Q12. Relations Case Study

$R = \{(x, y) : y = 3x\}$ check on Roll numbers $\{1, 2, \dots 30\}$.

1. List elements of R.
2. Check Reflexive, Symmetric, and Transitive properties.

Q13. Derivatives Case Study (Maxima/Minima)

Wooden box with square base (side x) and fixed volume (V). Minimize Surface Area (S).

1. Write S in terms of x and V.
2. Find dS/dx.
3. Find dimension relation for minimum area.

Quick Revision Keys (Quick Check)

- **Q2:** $|\text{adj } A| = |A|^{n-1} = 5^2 = 25$.
- **Q4:** Vectors are perpendicular (dot product = 0).
- **Q6:** Distance formula $d = |((b_1 \times b_2)\cdot(a_2 - a_1))| / |b_1 \times b_2|$.
- **Q11:** Apply $P(E_2|A) = [P(E_2)P(A|E_2)] / \Sigma P(E_i)P(A|E_i)$.